2025 International Conference on Applications of Machine Intelligence and Data Analytics (ICAMIDA) | 979-8-3315-2276-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICAMIDA64673.2025.11209758

Text Preprocessing Technique of Marathi Documents for Text Summarization

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***Abstract*—Text summarization is a process in which the original vast text is reduced into a smaller version while retaining its abstract meaning. The text summarizing is done in common global and regional languages like Marathi. Text summarization is an important topic in natural language processing (NLP), which aims to generate short and useful summaries from enormous amounts of text. The efficiency of summarization algorithms, however, is strongly dependent on the quality of the input data, making preprocessing an essential step. This work focuses on getting ready Marathi texts for text summarization, specifically addressing the challenges given by the Marathi language's unique linguistic traits. We propose a thorough preprocessing pipeline that comprises Marathi- specific tokenization, stop word removal, stemming, and POS Tagging. This work evaluates the proposed preprocessing methods on a given dataset of Marathi documents, focusing on how they affect the performance of extractive and abstractive summarization models.**

***Index Terms*—Natural Language Processing, Text Summarization, Preprocessing, Marathi Language**

I. Introduction

The fast rise of digital content has created an urgent demand for systems that can efficiently handle and condense large amounts of data. Text summarizing is important in this situation because it produces compact versions of papers while retaining the core concepts. Text summarization is utilized in a variety of applications, including news aggregation, content recommendation, and document retrieval, where users want concise, accurate summaries of vast materials. The quality of the summary, however, is determined not just by the summarization algorithms, but also by how effectively the material is prepared before to the summarization process. This makes text preprocessing a critical step in assuring the efficacy of any summarizing technique. While much of the study on text summarizing has focused on commonly spoken languages such as English, languages like Marathi have received less attention. Marathi, an Indo-Aryan language spoken largely in India's Maharashtra state, presents particular problems for natural language processing (NLP) applications such as text summarization [18]. Its advanced syntax, rich morphology, and regional vocabulary distinguish it from better researched languages. Standard preparation approaches designed for languages such as English struggle to operate efficiently on Marathi text due to these differences.

To improve Marathi text summarization, preparation approaches must be designed expressly to overcome these linguistic issues. The first step in text summarization is called preprocessing, and it involves a number of activities meant to get the text ready for more in-depth examination [10]. Tokenization (breaking down text into smaller units like words and phrases), part-of-speech tagging (determining each word's grammatical role), stopword removal (removing common words that don't add much meaning), and stemming (reducing words to their root forms) are important preprocessing tasks [6]. Although many NLP systems follow these processes, Marathi poses difficulties because of its complex grammar, word construction rules, and frequent usage of compound words. Furthermore, the challenge of creating reliable text is made more difficult by the absence of easily accessible, extensive datasets and Marathi-specific technologies. Tokenization is particularly challenging in Marathi since the language uses a lot of compound words, which are made up of two or more smaller words concatenated to form a larger, more complex word [18]. Accurate tokenization can be difficult because these compound words, which are frequently created by adding prefixes or suffixes to base words, can mask the meaning of the constituent words. Additionally, part-of- speech tagging in Marathi necessitates a thorough comprehension of the language's grammatical norms, which are very different from English's. When tagging words for their grammatical function, Marathi verbs, for example, use conjugations to indicate tense, aspect, and gender, which adds another level of complexity.

This work presents a unique preparation strategy to meet the issues of Marathi text summarization and employs Marathi-specific techniques like as tokenization, part-of- speech tagging, stopword removal, and stemming. Tokenization method provides for the development of compound words by breaking them down into meaningful components for further processing [11]. We also introduce a part-of-speech tagging technique tailored to the grammatical structure of Marathi. Furthermore, this includes a stopword list that reflects regional linguistic variances, guaranteeing that the preprocessing procedures can be applied to different forms of the language. Stemming techniques are intended to reduce words to their base forms,hence improving word matching and meaning extraction [17]. The purpose of this study is to show how these specialized preparation processes might improve the performance of Marathi text summarizing systems. By improving the quality of the input text, we hope to improve the accuracy and relevance of

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summaries generated by both extractive and abstractive summarizing models. Extractive summarizing selects key words or sentences straight from the text, whereas abstractive summarization creates new sentences that paraphrase the original content [4]. Both methods might benefit from better preprocessing to ensure that the summarization models work with clean, relevant data. This work highlights the necessity of language-specific preprocessing for low-resource languages such as Marathi, which require distinct approaches to manage their linguistic aspects.

II. Motivation

The purpose of this study is to speed up knowledge about NLPs techniques by providing an overview of current research, specifically in Text Summarization. It also makes possible to develop new resources, techniques, datasets, and tools to satisfy the demands of the industrial and research sectors. With the development of NLPs, automatic text summarization became practical for sentiment analysis and regular text document summaries. Additionally, text summarization encourages a flexible method of conducting research in a number of disciplines, including psychology, natural language, cognitive science, and machine learning. These combined results, which were obtained from a variety of sources, served as the inspiration for this study.

III. Literature Survey

Patil et.al [8] proposed a basic stemmer for the Marathi language that uses a suffix stripping technique and examines its impact, together with stop-word removal, on the Marathi text retrieval process. The results show that using the proposed suffix stripper and stop-word removal tool for Marathi text retrieval leads to considerable improvements in precision, r- precision, precision@10, and recall. Deshpande et.al [13] presented clustering results across multiple algorithms, and it is evident that the CCSCL K-means and GMM algorithms have shown similar performance in terms of cluster quality and performance metrics. The purpose of this study is to investigate how the proposed pre-processing methods affect the performance of the clustering algorithm. Document clustering was created using a variety of pre-processing approaches. Kakde et.al [11] studied to perform sentiment analysis supervised algorithms that is Multinomial Naıve Bayes is implemented on the Marathi dataset. Along with this deep learning, Natural Language Processing Approach Bidirectional Encoder Representations from Transformers (BERT) is utilized and fine-tuned for the specific work to evaluate more accuracy and State-of-the- Art results. The accuracy generated by the BERT language model bert-base-multilingual-uncased is 68.90 which is good compared to the supervised model. Jakhotiya et.al [1] stated that Text preprocessing is quite useful in structuring the data so as to design highly efficient models as per requirements. They indicate how using the various preprocessing techniques can have an impact on the accuracy level of the results targeted to be produced.Kadam et.al [18] stated that there has been a lot of work done on some Indian languages. For this reason, developing a single document text summarizer in Marathi is delicate. Their system can produce an abstract summary that, when abstracted can cut 44.48 percent of the original text. Their unborn pretensions include expanding the features and perfecting the styles for abstracting Marathi text. In addition, the database will be better to increase the summary text delicacy. Giri et.al [20] proposed a new stemmer for Marathi. It focuses on stripping suffixes to retrieve the root word form. The proposed stemmer uses a multilayer approach, taking into account both auxiliary verb- based and gender-based suffixes. The given approach aims to improve on the limitations of previously proposed stemmers for this language. MT Stemmer's stemming results in a considerable improvement in various performance measures. The proposed stemmer's functionality suggests that analogous engines for other Indic languages could be developed. The projected MT Stemmer is a positive step forward for the Marathi language.

Mrida et.al [12] provides a systematic survey of the broad ATS domain in different phases: performance measurement matrices, dataset inspections, important text summarization algorithms, feature extraction architectures, and fundamental theories with prior research backgrounds and challenges of current architectures. This work also outlines the existing drawbacks and difficulties with ATS techniques and algorithms, which should motivate researchers to work toward resolving these issues and overcoming recent changes and challenges. Tabassum et.al [7] emphasizes the need of applying text preparation techniques efficiently, as well as information retrieval using Natural Language Processing feature extraction methods. Natural language is difficult, and no machine learning model can achieve 100. Dhawale et.al [5] explored preprocessing methods for Marathi news articles. They found that the LINGO [Label Induction Grouping] algorithm can effectively improve Marathi document summarization outcomes. And also sentence scoring, ROUGE assessment, language-neutral syntax (LNS), support vector machine (SVM) classifier, and hybrid algorithms are appropriate for Indian languages. They have worked with NLP and information retrieval using a Python library. After forming the Key Value pair, they appended a list of words along with their frequency count. Ambedkar et.al [2] tried to survey various approaches to text summarizing that have been developed recently. They also discuss the salient features of several summarizing strategies. A few software applications used in legal text summarizing are briefly covered. At last, offer some suggestions for further investigation.

IV. Text Summarization

In NLP, text summarization is a difficult research task that yields a valuable summary of any given input document. That is, in fact, the process of reducing a long text document to a shorter version while maintaining the original meaning and overall sense[9]. A summary is a shorter version of an important text. The size of this summary text is consistently less than that of the main text.

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Fig. 1: Text Summarization

The primary task is to extract or choose the most important information from the original text, then represent that information to create a summary[18]. A summary is a short form of text document; it gives important information from the text document in condensed form in less time. The automatic text summarizer is needed in digital world to retrieve required information in fraction of seconds[11]. There are three ways to text summarization extractive, abstractive, and hybrid. Each approach has multiple approaches and procedures. Each approach has both advantages and downsides. Figure 2 provides an overview of the ways and specific methodologies used. There are three ways to text summarization extractive, abstractive, and Hybrid. Each approach has multiple approaches and procedures. Each approach has both advantages and downsides. A brief summary of the methodologies, together with. Figure 2 illustrates some specific ways.

A. Extractive Text Summarization

The Extractive text summarization approach seeks to find words and sentences in a text and use them efficiently to generate a summary [16]. This requires choosing sentences from the original document depending on its importance. These key sentences are then utilized to reproduce the text's core elements word for word, yielding a subset of the original document's phrases[14]. Extractive text summarization employs two types of machine learning approaches: supervised and unsupervised machine learning, as illustrated in Figure. The next section provides a quick summary of various sub classes.

- **Supervised Learning Methods:** In supervised learning approaches, the initial stage is to learn how to classify documents by training to recognize summarized and non-summarized texts [12]. These approaches' machine learning and neural network algorithms require a categorized dataset for training, which includes both summarized and non-summarized texts with lables.
- **Unsupervised Learning Methods:** It allows for summarization without human assistance, such as picking initial sentences from a document. These solutions just require complex algorithms, such as graph-based, concept- based, fuzzy logic, and latent semantics, to accept user input and work automatically. These approaches are useful for handling large amounts of data[14].

B. Abstractive Text Summarization

Abstractive text summarizing is the evolution and automation of the traditional approach of text summarization [17]. The abstractive process involves paraphrasing important sections

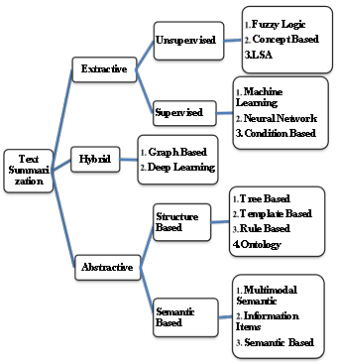

Fig. 2: Text Summarization approches

and main concepts from a literary piece. There are two sorts of abstractive summarization approaches: structure-based and semantic- based. A quick description of these two types based on natural language processing is provided below:

- **Structure Based Methods:** It filters crucial info from documents using abstract or cognitive algorithms. The algorithms for tree-based, template-based, and rule-based ontologies are the most widely used[14].
- **Semantic Based Methods:** This approach refines sentences by applying NLP to the entire manuscript. Some strategies can be used to readily find the noun and verb phrases. These methods include the multimodal semantic method (MSM), semantic graph-based method (SGM), information item-based method (IIM), and semantic text representation model (STRM)[12].

C. Hybrid Text Summarization

Hybrid text summarization is a combination of both extractive and abstractive text summarization.It has two types:

- **Graph Based Methods:** Both extractive and abstractive text summarization can be accomplished with the graph-based approach. Using a graph, this unsupervised learning technique assigns a rating to the necessary sentences or terms. It uses LexRank, PageRank, TextRank and many more techniques.
- **Deep Learning:** Information-driven ATS can be made more effective, accessible, and user-friendly with the use of deep learning models. Since these models aim to replicate human brain functions, they hold great promise for ATS. Since deep neural networks' architecture complements the complex structure of the language, they are frequently used to solve NLP problems.

V. Proposed Methodology

Following steps show the workflow for proposed system...

A. Data Collection

In this research work, we have taken 30 Marathi text documents which is considered as database for our research work. And performed all the preprocessing techniques steps on it. Marathi text documents to form a representative dataset. A

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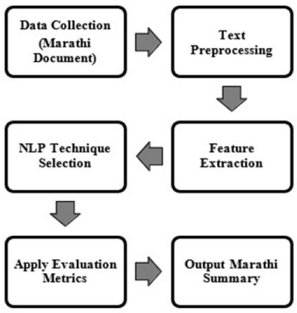

Fig. 3: Proposed Methodology Workflow

Marathi language text is accepted as an input for the proposed system. This is actually a single document source file.

B. Text Preprocessing

In this step we preprocess the Marathi text, including removal of punctuation, stemming, stop word removal, tokenization, lemmatization etc.

C. Feature Extraction

In natural language processing (NLP), feature extraction is a fundamental activity that includes turning raw text input into a format that machine learning algorithms can easily analyze. There are several strategies for feature extraction in NLP, each with advantages and disadvantages.

D. NLP Model Selection and Training

After that we can select any NLP techniques like rule based, computational statistics to perform further processing on that data. Select a suitable NLP model for Marathi. Train the model on the preprocessed dataset for summarization.

E. Implement Evaluation Metrics

In this step, we implement evaluation metrics to assess summary quality and iteratively refine the model based on evaluations. And also checks that how the model is performing for getting accuracy in results.

F. Output as Marathi Summary Generation

And at the end,a short and precise summary of document will be generated as our result and output.

VI. Text Preprocessing

Text pre-processing is the process of transforming unstructured and noisy text data into a clean and consistent format. Text preprocessing is an important stage in Natural Language Processing (NLP) since it prepares raw text input for subsequent analysis, such as text categorization, named entity identification, sentiment analysis, or machine translation [19]. Preprocessing procedures for Marathi text are comparable to those used for other languages, although they frequently require additional processes to address Marathi's distinct qualities, such as script, morphology, and syntactic structure [15].

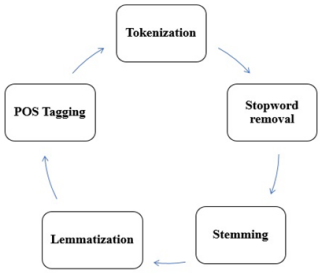

Fig. 4: Text Preprocessing Steps

A. Text Normalization

Marathi text, like other languages, may have informal terms, regional variants, or uneven spelling. Remove unnecessary non-alphanumeric characters, symbols, and punctuation [1]. To repair misspelled words, use a dictionary or spell checker.

B. Tokenization

To examine text structure, break it into smaller pieces (e.g. words or phrases) by tokenization. For Marathi, this includes: Word tokenization involves breaking down text into individual words, including compound words (common in Marathi). Tokenization involves breaking down material into distinct sentences for processing [3]. Tokenizers such as stanza and spaCy can handle this, particularly with pre- trained models.

C. Stopword Removal

Stopwords are common words that appear frequently but do not contribute significantly to the meaning or interpretation of the text[3]. To filter out certain terms, utilize a prepared list of Marathi stopwords. Many NLP technologies include stopword lists for Marathi.

D. Stemming

Stemming is the process of removing suffixes from words to reveal their root form[3]. While stemming is extensively employed in English and other languages, it is less popular in Marathi than lemmatization because it can oversimplify complicated word forms. Use a Marathi stemmer to remove common suffixes.

E. Lemmatization

Lemmatization reduces words to their root form. This is especially significant in languages such as Marathi, which have complicated word structures due to suffixes, inflections, and conjugations [14]. Lemmatization guarantees that all of these variants are regarded as one word.

F. POS Tagging (Parts of Speech Tagging

Identifying the grammatical role of each word in a phrase aid in comprehending sentence structure, extracting features for machine learning, and performing other tasks[14]. POS tagging assigns labels such as noun (NOUN), verb (VERB), adjective (ADJ), pronoun (PRON), and so on.

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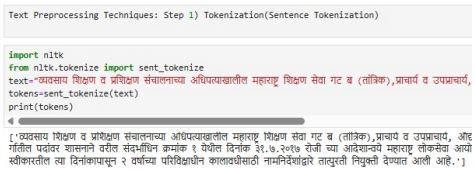

Fig. 5: Python Code for Tokenization

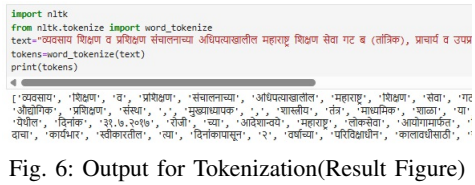

Fig. 6: Output for Tokenization(Result Figure)

VII. Results

This study effectively used Marathi text preprocessing techniques such as tokenization, stopword removal, stemming, lemmatization, and POS tagging. Sentence tokenization with NLTK correctly recognized sentence boundaries, ensuring grammatical integrity. Stopword elimination significantly decreased unnecessary words, resulting in increased text efficiency. Stemming and lemmatization were compared, and lemmatization proved to be more effective at dealing with complicated word structures. POS tagging using NLP models like Stanza correctly detected grammatical roles, which improved syntactic representation. These findings support the usefulness of the methodologies used, giving a solid foundation for advanced NLP applications in Marathi text analysis. The above figures Fig. 5 and Fig. 6 shows the tokenization step using Python programming along with its output. We have taken only two snapshots of the code as a sample among all the implemented code.

VIII. Acknowledgment

My sincere gratitude to Dr. Sonal S. Deshmukh, Head of Department, Master of Computer Applications, MGM's JNEC Chhatrapati Sambhajinagar, for their constant support throughout this work. They have given me the inspiration, drive, and support I needed.

IX. Conclusion

This study presents a preprocessing specifically for Marathi text to improve text summarization. Marathi has unique issues such as compound words, complex syntax, and regional language variances that traditional preprocessing approaches cannot handle effectively. Tokenization, part-of-speech tagging, stopword elimination, and stemming are all crucial aspects in our technique, which is specifically customized to Marathi. We ensure that compound words are effectively tokenized, grammar is properly marked, and relevant names are identified. By deleting redundant words and reducing them to their core forms, we enable summarization models to focus on vital content. The suggested approach enhances the quality of Marathi text for summarization models, resulting in more accurate summaries.

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